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Paper 1 of 1 — CHE 510 CHEMICAL REACTION ENGINEERING ii

Level 500 • First Semester • Handout

CHE 510

CHEMICAL REACTION
ENGINEERING II

© Page 1

CHE 510 CHEMICAL REACTION ENGINEERING II

TABLE OF CONTENTS

CHAPTER 1: TEMPERATURE AND PRESSURE EFFECTS

- Heats of Reaction from Thermodynamics
- Equilibrium Constants from Thermodynamics
- Optimum temperature progression, adiabatic operations. Reactor design calculations under non – isothermal conditions

CHAPTER 2: DISTRIBUTIONS OF RESIDENCE TIMES FOR CHEMICAL REACTORS

- Residence Time Distribution (RTD)
- Measurement of the RTD
- Characteristics of the RTD
- Reactor Modeling Using the RTD
- Basics of Nonideal Flow
- Zero-Parameter Models
- One-Parameter Models for Nonideal Flow
- Two-Parameter Models for Nonideal Flow
- Terminology

CHAPTER 3: SUMMARY OF CATALYTIC AND NON-CATALYTIC SYSTEMS

- Reactions Catalyzed by Solids
- Non-catalytic Systems

CHAPTER 4: MULTIPLE REACTIONS

- Multiple reactions: product distribution and temperature
- Reactor Design for Multiple Reactions

© Page 2

TEMPERATURE AND PRESSURE EFFECTS ON REACTIONS

In our search for favorable conditions for reaction we have considered how reactor type and size influence the extent of conversion and distribution of products. The reaction temperature and pressure also influence the progress of reactions, and it is the role of these variables that we now consider.

We follow a three-step procedure:

- First, we must find how equilibrium composition, rate of reaction, and product distribution are affected by changes in operating temperatures and pressures. This

will allow us to determine the optimum temperature progression, and it is this that we strive to approximate with a real design.

□ Second, chemical reactions are usually accompanied by heat effects, and we must know how these will change the temperature of the reacting mixture. With this information we are able to propose a number of favorable reactor and heat exchange systems-those which closely approach the optimum.

□ Finally, economic considerations will select one of these favorable systems as the best.

With single reactions we are concerned with conversion level and reactor stability. Questions of product distribution do not occur.

Thermodynamics gives two important pieces of information, the first being the heat liberated or absorbed for a given extent of reaction, the second being the maximum possible conversion.

1.1 Heats of Reaction from Thermodynamics

The heat liberated or absorbed during reaction at temperature T_2 depends on the nature of the reacting system, the amount of material reacting, and the temperature and pressure of the reacting system, and is calculated from the heat of reaction, for the reaction in question. When this is not known, it can in most cases be calculated from known and tabulated thermochemical data on heats of formation, or heats of combustion, of the reacting materials.

These are tabulated at some standard temperature, T_1 , usually 25°C . As a brief reminder, consider the reaction

By convention we define the heat of reaction at temperature T as the heat transferred to the reacting system from the surroundings when a moles of A disappear to produce r moles of R

© Page 3

and s moles of S with the system measured at the same temperature and pressure before and after the change. Thus

→ {

1.2 Heats of Reaction and Temperature

The first problem is to evaluate the heat of reaction at temperature T_2 knowing the heat of reaction at temperature T_1 . This is found by the law of conservation of energy as follows:

In terms of enthalpies of reactants and products this becomes

where subscripts 1 and 2 refer to quantities measured at temperatures T_1 and T_2 , respectively. In terms of specific heats

where

When the molar specific heats are functions of temperatures as follows,

We obtain

© Page 4

where

Knowing the heat of reaction at any one temperature as well as the specific heats of reactants and products in the temperature range concerned allows us to calculate the heat of reaction at any other temperature. From this the heat effects of the reaction can be found.

EXAMPLE: AT VARIOUS TEMPERATURES

SOLUTION

© Page 5

1.3 Equilibrium Constants from Thermodynamics

From the second law of thermodynamics equilibrium constants, hence equilibrium compositions of reacting systems, may be calculated. We must remember, however, that real systems do not necessarily achieve this conversion; therefore, the conversions calculated from thermodynamics are only suggested attainable values.

As a brief reminder, the standard free energy for the reaction of Eq. 1 at temperature T is defined as

where G^0 is the standard free energy of a reacting component, tabulated for many compounds; and K is the thermodynamic equilibrium constant for the reaction.

Equilibrium Conversion: The equilibrium composition, as governed by the equilibrium constant, changes with temperature, and from thermodynamics the rate of change is given by

© Page 6

On integrating Eq. 15, we see how the equilibrium constant changes with temperature.

When the heat of reaction, ΔH^0 , can be considered to be constant in the temperature interval, integration yields (Van Hoff's equation for)

The following conclusions may be drawn from thermodynamics. These are illustrated in part by Fig. 9.1.

Figure 9.1 Effect of temperature on equilibrium conversion as predicted by thermodynamics (pressure fixed.)

1. The thermodynamic equilibrium constant is unaffected by the pressure of the system, by the presence or absence of inerts, or by the kinetics of the reaction, but is affected by the temperature of the system.
2. Though the thermodynamic equilibrium constant is unaffected by pressure or inerts, the equilibrium concentration of materials and equilibrium conversion of reactants can be influenced by these variables.
3. $K \gg 1$ indicates that practically complete conversion may be possible and that the reaction can be considered to be irreversible. $K \ll 1$ indicates that reaction will not proceed to any appreciable extent.
4. For an increase in temperature, equilibrium conversion rises for endothermic reactions and drops for exothermic reactions.

© Page 7

EXAMPLE: EQUILIBRIUM CONVERSION AT DIFFERENT TEMPERATURES

SOLUTION

© Page 8

© Page 9

Figure E9.2

NONISOTHERMAL REACTORS

Although isothermal conditions are most useful for the measurement of kinetic data, real reactor operation is normally nonisothermal. Within the limits of heat exchange, the reactor can operate isothermally (maximum heat exchange) or adiabatically (no heat exchange). Between these bounds of heat transfer lies the most common form of reactor operation – the nonisothermal regime (some extent of heat exchange). The three types of reactor operations yield different temperature profiles within the reactor operation and are illustrated in Figure 4.1 for an exothermic reaction

© Page 10

Figure 4.1: Temperature profiles in a PFR accomplishing an exothermic reaction

If a reactor is operated at nonisothermal or adiabatic conditions then the material balance equation must be written with the temperature, T , as a variable. For example with the PFR, the material balance becomes:

Since the reaction rate expression now contains the independent variable T , the material balance cannot be solved alone. The solution of the material balance equation is only possible by the simultaneous solution of the energy balance. Thus, for nonisothermal reactor descriptions, an energy balance must accompany the material balance.

Due to the exponential dependence of the reaction rate constant on the temperature, the coupling of mass and energy balance equations usually requires the solution of simultaneous non-linear differential equation.

76.4 HEAT EFFECTS

© Page 11

© Page 12

© Page 13

© Page 14

Figure 76.3 Algorithm for nonisothermal CSTR design. (Source: Fogler, H. S. 1992. The Elements of Chemical Reaction Engineering, 2nd ed. Prentice Hall, Englewood Cliffs, NJ.)

© Page 15

Figure 8-13 Algorithm for adiabatic CSTR design

© Page 16

TABLE8- 4. WAYS TO SPECIFY THE SIZING A CSTR

© Page 17

© Page 18

© Page 19

© Page 20

DISTRIBUTIONS OF RESIDENCE TIMES FOR CHEMICAL REACTORS

2.1 RESIDENCE TIME DISTRIBUTION (RTD)

As its name implies, ideal flow is a model of flow which, in one of its various forms, may be closely approached, but is not actually achieved. Examples include the backmix flow (BMF) and plug flow (PF). The plug flow and continuous stirred tank models represent the two extremes in idealised flow regimes. The plug flow regime assumes total segregation between fluid elements, while the CSTR model represents total and instantaneous mixing.

Although these two flow behaviour can be easily described, flow fields that deviate from these limits are extremely complex and become impractical to completely model. However, it is often not necessary to know the details of the entire flow field but rather only how long fluid elements reside in the reactor (i.e. the distribution of residence times). This information can be used as a diagnostic tool to ascertain flow characteristics of a particular reactor.

The "age" of a fluid element is defined as the time it has resided within the reactor. The concept of a fluid element being a small volume relative to the size of the reactor yet sufficiently large to exhibit continuous properties such as density and concentration was first put forth by Danckwerts in 1953. Consider the following experiment: a tracer (could be a particular chemical or radioactive species) is injected into a reactor, and the outlet stream is monitored as a function of time. The results of these experiments for an ideal PFR and CSTR are illustrated in Figure 2.1

If an impulse is injected into a PFR, an impulse will appear in the outlet because there is no fluid mixing. The pulse will appear at a time $t_1 = t_0 + \tau$, where T is the space time ($T = V/v$). However, with the CSTR, the pulse emerges as an exponential decay in tracer concentration, since there is an exponential distribution in residence times. For all nonideal reactors, the results must lie between these two limiting cases.

In order to analyze the residence time distribution of the fluid in a reactor the following relationships have been developed. Fluid elements may require differing lengths of time to travel through the reactor. The distribution of the exit times, defined as the $E(t)$ curve, is the residence time distribution (RTD) of the fluid. The exit concentration of a tracer species $C(t)$ can be used to define $E(t)$. That is:

$$\int_0^\infty E(t) dt = 1$$

Such that:

$$\int_0^\infty C(t) dt = \frac{V}{v}$$

© Page 21

Figure 2.1: Concentrations of tracer species using an impulse input. (a) PFR ($t_1 = t_0 + \tau$). (b) CSTR. (c) Nonideal reactor.

© Page 22

With this definition, the fraction of the exit stream that has residence time (age) between t and $t + dt$ is:

while the fraction of fluid in the exit stream with age less than t_1 is:

∫

In a non-ideal system, the distribution functions exhibit various forms. Therefore, knowledge of these distributions for a given system is important. By comparing the curves corresponding to ideal systems, it is possible to determine the type of ideal behavior that a system most closely resembles. If the differences are small, the system can be treated as one of the ideal models. However, a large difference may imply a complex system, in which case modification will be required to reflect the real behavior of the system. The techniques for measuring the RTD by means of a tracer offer a diagnostic approach in analyzing the structure of flow in continuous systems.

© Page 23

2.2 MEASUREMENT OF THE RTD

The RTD is determined experimentally by injecting an inert chemical, molecule or atom, called a tracer, into the reactor at some time $t = 0$ and then measuring the tracer concentration, C , in the effluent stream as a function of time. In addition to being a nonreactive species that is easily detectable, the tracer should have physical properties similar to those of the reacting mixture and be completely soluble in the mixture. It also should not adsorb on the walls or other surfaces in the reactor. The latter requirements are needed so that the tracer's behavior will honestly reflect that of the material flowing through the reactor. Colored and radioactive materials along with inert gases are the most common types of tracers. The two most used methods of injection are pulse input and step input which we will consider in this section. Others include the periodic and random input.

2.2.1 Pulse Input Experiment

A pulse input involves the rapid injection of a small, known amount of tracer, as a physician might use a syringe to inject medication. In a pulse input, an amount of tracer N_0 is suddenly injected in one shot into the feedstream entering the reactor in as short a time as possible. The outlet concentration is then measured as a function of time. Typical concentration-time curves at the inlet and outlet of an arbitrary reactor are shown in Fig 2.2. The effluent concentration-time curve is referred to as the C curve in RTD analysis. We shall analyze the injection of a tracer pulse for a single-input and single-output system in which only flow (i.e., no dispersion) carries the tracer material across system boundaries.

First, we choose an increment of time sufficiently small that the concentration of tracer, $C(t)$, exiting between time t and $t + dt$ is essentially the same. The amount of tracer material, dN , leaving the reactor between time t and $t + dt$ is then

where v is the effluent volumetric flow rate. In other words, dN is the amount of material exiting the reactor that has spent an amount of time between t and $t + dt$ in the reactor. If

we now divide by the total amount of material that was injected into the reactor, NO, we obtain

which represents the fraction of material that has a residence time in the reactor between time t and $t + \Delta t$.

For pulse injection we define

© Page 24

So that

The quantity $E(t)$ is called the residence-time distribution function. It is the function that describes in a quantitative manner how much time different fluid elements have spent in the reactor. The quantity $E(t)dt$ is the fraction of fluid exiting the reactor that has spent between time t and $t + dt$ inside the reactor.

Fig 2.2: RTD measurements

© Page 25

2.2.2 Step Tracer Experiment

A step-change input involves an instantaneous change in inlet concentration of tracer from an initial steady-state level (which may be zero) to a second level.

© Page 26

© Page 27

© Page 28

© Page 29

2.3 CHARACTERISTICS OF THE RTD

Sometimes $E(t)$ is called the exit-age distribution function. If we regard the "age" of an atom as the time it has resided in the reaction environment, then $E(t)$ concerns the age distribution of the effluent stream. It is the most used of the distribution functions connected with reactor analysis because it characterizes the lengths of time various atoms spend at reaction conditions

2.3.1 Integral Relationships

The fraction of the exit stream that has resided in the reactor for a period of time shorter than a given value t is equal to the sum over all times less than t of $E(t)$, or expressed continuously.

$F(t)$

Where $F(t)$ is the cumulative RTD function

Analogously, we have

© Page 30

$f(t)$

Figure 2-3 Cumulative distribution curve. $F(t)$ (The F-curve)

2.3.2 Other Moments of the RTD

It is very common to compare RTDs by using their moments instead of trying to compare their entire distributions. For this purpose, three moments are normally used.

□ The first is the mean residence time.

□ The second moment commonly used is taken about the mean and is called the variance, or square of the standard deviation. It is defined by

The magnitude of this moment is an indication of the "spread" of the distribution; the greater the value of this moment is, the greater a distribution's spread will be.

□ The third moment is also taken about the mean and is related to the skewness. The skewness is defined by

© Page 31

The magnitude of this moment measures the extent that a distribution is skewed in one direction or another in reference to the mean.

Rigorously, for complete description of a distribution, all moments must be determined. Practically, these three are usually sufficient for a reasonable characterization of an RTD.

PART 2 PREDICTING CONVERSION AND EXIT CONCENTRATION

2.4 REACTOR MODELING USING THE RTD

Now that we have characterized our reactor and have gone to the lab to take data to determine the reaction kinetics, we need to choose a model to predict conversion in our real reactor.

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We now present the five models shown in Table 2-1. We shall classify each model according to the number of adjustable parameters. Ways we use the RTD data to predict conversion in nonideal reactors

TABLE 2.1 MODELS FOR PREDICTING CONVERSION FROM RTD DATA

Model Classification Model Types

Zero adjustable parameters Segregation model

Maximum mixedness model

One adjustable parameter Tank-in-series model

Dispersion model

Two adjustable parameters Real reactors modelled as combinations of ideal reactors.

The tank-in-series (TIS) and the dispersion plug flow (DPF) models can be adopted as reactor models once their parameters (e.g., Number of tanks, N , Dispersion coefficient, Da and Peclet number, NPe) are known. However, these are macromixing models, which are unable to account for non-ideal mixing behavior at the microscopic level. This section reviews two micromixing models for evaluating the performance of a reactor— the segregated flow model and the maximum mixedness model—and considers the effect of micromixing on conversion.

© Page 32

2.5 BASICS OF NON-IDEAL FLOW

Three main factors affect either the interaction of fluids in a system or the flow pattern.

These are:

- How early or late mixing of material occurs in the system.
- The state of aggregation of the flowing material, its tendency to clump, and for a group of molecules to move about together.
- The residence time distribution of material that is flowing through the system.

The fluid elements of a single flowing stream can mix with each other either early or late in their flow through the system. This is especially important with two reactants entering a vessel. Two extreme states of aggregation of a process are referred to as macrofluid and microfluid (Figure 2-4). In a macrofluid, molecules move together in clumps. The clumps are distributed in residence times, but all the molecules within a clump have the same age. In contrast, the clumps are dispersed in a microfluid because all the molecules move independently of one another regardless of age.

The RTD in a system is a measure of the degree to which fluid elements mix. In an ideal plug flow reactor, there is no mixing, while in a perfect mixer, the elements of different ages are uniformly mixed. A real process fluid is neither a macrofluid nor a microfluid, but tends toward one or the other of these extremes. Fluid mixing in a vessel is a complex process and can be analysed on both macroscopic and microscopic scales. In a non-ideal system, there are irregularities that account for the fluid mixing of different ages at the microscopic level. The RTD does not account for how fluid elements at the microscopic scale interact with each other. There is an obvious distinction between macromixing, or a set of mechanisms associated with the RTD, and micromixing, which results from the interactions between the fluid elements.

Macromixing defines the variation in the residence time experienced by molecules flowing through a flow system, while micromixing relates to the concentration history experienced by the molecules as they traverse through the system. The conversion in a system can be determined from the RTD with knowledge of the macromixing process.

© Page 33

Figure 2-4. Two extreme states of fluid micromixing.

Generally, the performance of non-ideal flow systems can be determined from known parameters, such as:

- The kinetics of the system
- The RTD of fluid in the system
- The earliness or lateness of fluid mixing in the system
- The nature of the fluid (i.e., whether a micro or macro fluid) in the system

NOTE:

Segregation model mixing occurs at the latest possible point while Maximum mixedness mixing occurs at the earliest possible point.

© Page 34

2.6 ZERO-PARAMETER MODELS

2.6.1 Segregation Model

In a "perfectly mixed" CSTR, the entering fluid is assumed to be distributed immediately and evenly throughout the reacting mixture. This mixing is assumed to take place even on the microscale, and elements of different ages mix together thoroughly to form a completely micromixed Fluid. If fluid elements of different ages do not mix together at all, the elements remain segregated from each other, and the fluid is termed completely segregated. The extremes of complete micromixing and complete segregation are the limits of the micromixing of a reacting mixture.

In developing the segregated mixing model, we first consider a CSTR because the application of the concepts of mixing quality are illustrated most easily using this reactor type. In the segregated flow model we visualize the flow through the reactor to consist of a continuous series of globules (Figure 2-5).

Figure 2-5 Little batch reactors (globules) inside a CSTR.

These globules retain their identity; that is, they do not interchange material with other globules in the fluid during their period of residence in the reaction environment. i.e., they remain segregated. In addition, each globule spends a different amount of time in the reactor. In essence, what we are doing is lumping all the molecules that have exactly the same residence time in the reactor into the same globule. The principles of reactor performance in the presence of completely segregated mixing were first described by Danckwerts and Zwietering.

Another way of looking at the segregation model for a continuous flow system is the PFR shown in Figures 2-6(a) and (b). Because the fluid flows down the reactor in plug flow, each exit stream corresponds to a specific residence time in the reactor. Batches of molecules are removed from the reactor at different locations along the reactor in such a manner as to

© Page 35

duplicate the RTD function $E(t)$. The molecules removed near the entrance to the reactor correspond to those molecules having short residence times in the reactor.

Figure 2-6 Mixing at the latest possible point.

Physically, this effluent would correspond to the molecules that channel rapidly through the reactor. The farther the molecules travel along the reactor before being removed, the longer their residence time. The points at which the various groups or batches of molecules are removed correspond to the RTD function for the reactor.

Because there is no molecular interchange between globules, each acts essentially as its own batch reactor. The reaction time in any one of these tiny batch reactors is equal to the time that the particular globule spends in the reaction environment. The distribution of residence times among the globules is given by the RTD of the particular reactor.

© Page 36

2.6.2 Maximum Mixedness Model

In a reactor with a segregated fluid, mixing between panicles of fluid does not occur until the fluid leaves the reactor. The reactor exit is, of course, the [arrest possible point that mixing can occur, and any effect of mixing is postponed until after all reaction has taken place as shown in Figure 2-6. We can also think of completely segregated flow as being in a state of minimum mixedness. We now want to consider the other extreme, that of maximum mixedness consistent with a given residence-time distribution.

We return again to the plug-flow reactor with side entrances, only this time the fluid enters the reactor along its length (Figure 2-7). As soon as the fluid enters the reactor, it is completely mixed radially (but not longitudinally) with the other fluid already in the reactor. The entering fluid is fed into the reactor through the side entrances in such a manner that the RTD of the plug-flow reactor with side entrances is identical to the RTD of the real reactor.

Figure 2-7 Mixing at the earliest possible point.

The globules at the far left of Figure 2-7 correspond to the molecules that spend a long time in the reactor while those at the far right correspond to the molecules that channel through the reactor. In the reactor with side entrances, mixing occurs at the earliest possible moment consistent with the RTD. Thus, the effect of mixing occurs as early as possible throughout the reactor, and this situation is termed the condition of maximum mixedness

© Page 37

2.7 ONE-PARAMETER MODELS

Here we use a single parameter to account for the nonideality of our reactor. This parameter is most always evaluated by analyzing the RTD determined from a tracer test.

Examples of one-parameter models for nonideal CSTRs include a reactor dead volume V_d , where no reaction takes place, or a fraction f of fluid bypassing the reactor, thereby exiting unreacted.

Examples of one-parameter models for tubular reactors include the tanks-in-series model and the dispersion model. For the tanks-in-series model, the parameter is the number of tanks, N , and for the dispersion model, it is the dispersion coefficient, Da . Knowing the parameter values, we then proceed to determine the conversion and/or effluent concentrations for the reactor.

2.7.1 Tanks-in-Series (T-I-S) Model

In this section we discuss the use of the tanks-in-series (T-I-S) model to describe nonideal reactors and calculate conversion. The T-I-S model is a one-parameter model. We will analyze the RTD to determine the number of ideal tanks, n , in series that will give approximately the same RTD as the nonideal reactor. Next we will apply the reaction engineering algorithm to calculate conversion. We are first going to develop the RTD equation for three tanks in series (Figure 2-8) and then generalize to N reactors in series to derive an equation that gives the number of tanks in series that best fits the RTD data. Note

that tanks in series can approximate a PFR.

© Page 38

Figure 2-8 Tanks in series: (a) real system. (b) model system

2.7.2 Dispersion Model

The dispersion model is also used to describe nonideal tubular reactors. In this model, there is an axial dispersion of the material, which is governed by an analogy to Fick's law of diffusion, superimposed on the flow as shown in Figure 2.9. So in addition to transport by bulk flow, UACC, every component in the mixture is transported through any cross section of the reactor at a rate equal to $[-DaAC(dC/dz)]$ resulting from molecular and convective diffusion. By convective diffusion (i.e., dispersion) we mean either Aris-Taylor dispersion in laminar flow reactors or turbulent diffusion resulting from turbulent eddies.

Radial concentration profiles for plug flow (a) and a representative axial and radial profile for dispersive flow (b) are shown in Figure 2.9. Some molecules will diffuse forward ahead of molar average velocity while others will lag behind.

Figure 2-9 Concentration profiles (a) without and (b) with dispersion.

© Page 39

2.8 TWO-PARAMETER MODELS – MODELING REAL REACTORS WITH COMBINATION OF IDEAL REACTORS

2.8.1 Real CSTR Modeled Using Bypassing and Dead Space

A real CSTR is believed to be modeled as a combination of an ideal CSTR of volume V_s , a dead zone of volume V_d , and a bypass with a volumetric flow rate v_b (Figure 2-10). We have used a tracer experiment to evaluate the parameters of the model V_s and v_s . Because the total volume and volumetric flow rate are known, once V_s and v_s are found, v_b and V_d can readily be calculated.

Figure 2-10 (a) Real system: (b) model system.

2.8.2 Real CSTR Modeled as Two CSTRs with Interchange

In this particular model there is a highly agitated region in the vicinity of the impeller; outside this region, there is a region with less agitation (Figure 2-11). There is considerable material transfer between the two regions. Both inlet and outlet flow channels connect the highly agitated region. We shall model the highly agitated region as one CSTR, the quieter region as another CSTR with material transfer between the two.

© Page 40

Figure 2-11 (a) Real reaction system; (b) model reaction system

2.8.3 Other Models of Nonideal Reactors Using CSTRs and PFRs

Several reactor models have been discussed in the preceding pages. All are based on the physical observation that in almost all agitated tank reactors, there is a well-mixed zone in the vicinity of the agitator. This zone is usually represented by a CSTR. The region outside this well-mixed zone may then be modeled in various fashions. We have already considered

the simplest models, which have the main CSTR combined with a dead-space volume: if some short-circuiting of the feed to the outlet is suspected, a bypass stream can be added. The next step is to look at all possible combinations that we can use to model a nonideal reactor using only CSTRs, PFRs, dead volume, and bypassing. The rate of transfer between the two reactors is one of the model parameters. The positions of the inlet and outlet to the model reactor system depend on the physical layout of the real reactor.

Figure 2-12(a) describes a real PFR or PBR with channeling that is modeled as two PFRs/PBRs in parallel. The two parameters are the fraction of flow to the reactors [i.e. β and $(1 - \beta)$] and the fractional volume [i.e. α and $(1 - \alpha)$] of each reactor. Figure 2-12(b) describes a real PFR/PBR that has a backmix region and is modeled as a PFR/PBR in parallel with a CSTR. Figures 2-12(a) and (b) show a real CSTR modeled as two CSTRs with interchange. In one case, the fluid exits from the top CSTR (a) and in the other case the fluid exits from the bottom CSTR. The parameter β represents the interchange volumetric flow rate and α the fractional volume of the top reactor, where the fluid exits the reaction system. We note that the reactor in model 2-12(b) was found to describe extremely well a real reactor used in the production of terephthalic acid. A number of other combinations of ideal reactors can be found in Levenspiel.

© Page 41

Figure 2-12: Combinations of ideal reactors used to model real tubular reactors.

(a) two ideal PFRs in parallel (b) ideal PFR and Ideal CSTR in parallel

© Page 42

2.9 TERMINOLOGY

- Element of fluid: an amount of fluid small with respect to vessel size, but large with respect to molecular size, such that it can be characterized by values of (macroscopic) properties such as temperature, pressure, density and concentration.
- Age of a fluid element: length of time, from entry, that an element of fluid has been in a vessel at a particular instant. The age is equal to the residence time for those molecules that are just leaving the vessel.
- Dispersion: The movement of aggregates of molecules under the influence of a gradient of concentration, temperature, etc. The effect is represented by Fick's law with a dispersion coefficient substituted for molecular diffusion coefficient.
- Impulse: An amount of tracer injected instantaneously into a vessel at time zero.
- Macromixing: The phenomenon whereby residence times of clumps are distributed about a mean value. Mixing on a scale greater than the minimum striation thickness, by laminar or turbulent motion.
- Maximum mixedness: Exists when any molecule that enters a vessel immediately becomes associated with those molecules with which it will eventually leave the vessel. This occurs with those molecules that have the same life expectation. A state of maximum mixedness is associated with every residence time distribution (RTD).
- Maximum mixedness model: The fluid in a flow reactor that behaves as a micro fluid.

Mixing of molecules of different ages occurs as early as possible.

□ Mean residence time: The average time spent by the molecules in a vessel. It is the first moment of the effluent concentration from a vessel with impulse input and is defined by

$$\int$$

$$\int$$

□ Micromixing: Mixing among molecule of different ages (i.e., mixing between macrofluid clumps). Mixing on a scale smaller than the minimum eddy size or minimum striation thickness by molecular diffusion.

□ Mixing, ideal or complete: A state of complete uniformity of composition and temperature in a vessel. In a flow system, the residence time distribution is exponential, ranging from zero to infinity.

□ Pulse: A type of input in which the concentration of tracer in the input stream is changed suddenly, maintained at a non-zero value for a definite period, then changed to the original value and maintained that way for the period of interest. When the pulse is of constant magnitude it is called a square pulse.

□ Residence time (of an element of fluid): length of time that an element of fluid spends in a vessel (i.e., from entry to exit); a residence time is an age,

□ Residence time distribution (RTD): The time that elapses from when the element enters the vessel to when it leaves.

© Page 43

□ Residence time distribution functions: These give information about the fraction of the fluid that spends a certain time in a process vessel.

In the case of tracer from a vessel that contained an initial average concentration C_0 , the area under a plot of between the ordinates at t_1 and t_2 is the fraction of the molecules that have residence times in this range.

In the case of step constant input of concentration C_f to a vessel with zero initial concentration, the ratio at t_1 is the fraction of molecules with residence time less than t_1 .

□ Tracer: A substance that is used for measuring the residence time distribution in a vessel. Usually, it is inert and used in small concentrations so as not to change the physical properties of the process fluid appreciably, and analysable for accuracy.

□ Segregation: The tendency of the contents of a reactor to behave as a macrofluid.

□ Segregated flow: Occurs when all molecules that enter together also leave together. A state of segregation is associated with every RTD. Each aggregate of molecules reacts independently of every other aggregate, as in an individual batch reactor.

□ Segregated flow model: The fluid in a flow reactor is assumed to behave as a macrofluid. Each clump functions as a miniature batch reactor. Mixing of molecules of different ages occurs as late as possible.

□ Step Input: An input in which the concentration of tracer is changed to some constant value C_f at time zero and maintained at this level indefinitely. For example,

$C_{fu}(t-a)$ represents a step of magnitude C_f beginning at $t=a$. The resulting effluent concentration is C_U .

© Page 44

REACTIONS CATALYZED BY SOLIDS (LEVENSPIEL RXN TEXTBOOK)

Chapter 17 Heterogeneous Reactions-Introduction

Chapter 18 Solid Catalyzed Reactions

Chapter 19 The Packed Bed Catalytic Reactor

Chapter 20 Reactors with Suspended Solid Catalyst, Fluidized Reactors of Various Types

Chapter 21 Deactivating Catalysts

Chapter 22 G/L Reactions on Solid Catalyst: Trickle Beds, Slurry Reactors, Three Phase Fluidized Beds

CHAPTER 17 HETEROGENEOUS REACTIONS-INTRODUCTION

This section treats the kinetics and design of chemical reactors for heterogeneous systems of various kinds, each chapter considering a different system. For these systems there are two complicating factors that must be accounted for beyond what is normally considered in homogeneous systems. First, we have the complication of the rate expression, and second the complication of the contacting patterns for two-phase systems. Let us briefly discuss these in turn.

The Complications of the Rate Equation

Since more than one phase is present, the movement of material from phase to phase must be considered in the rate equation. Thus the rate expression in general will incorporate mass transfer terms in addition to the usual chemical kinetics term. These mass transfer (diffusion) terms are different in type and numbers in the different kinds of heterogeneous systems; hence, no single rate expression has general application. Here are some simple examples.

Contacting Patterns for Two-Phase Systems

There are many ways that two phases can be contacted, and for each the design equation will be unique. Design equations for these ideal flow patterns may be developed without too much difficulty. However, when real flow deviates considerably from these, we can do one of two things: we may develop models to mirror actual flow closely, or we may calculate performance with ideal patterns which "bracket" actual flow. Fortunately, most real reactors

© Page 45

for heterogeneous systems can be satisfactorily approximated by one of the five ideal flow patterns of Fig. 17.1. Notable exceptions are the reactions which take place in fluidized beds. There special models must be developed.

Figure 17.1. Ideal contacting patterns for two flowing fluids.

CHAPTER 18 SOLID CATALYZED REACTIONS

With many reactions, the rates are affected by materials which are neither reactants nor products. Such materials called catalysts can speed a reaction by a factor of a million or much more, or they may slow a reaction (negative catalyst).

There are two broad classes of catalysts: those that operate at close to ambient temperature with biochemical systems, and the man-made catalysts that operate at high temperature.

The biochemical catalysts, called enzymes, are found everywhere in the biochemical world and in living creatures, and without their action I doubt that life could exist at all.

The man-made catalysts, mostly solids, usually aim to cause the high-temperature rupture or synthesis of materials. These reactions play an important role in many industrial processes, such as the production of methanol, sulfuric acid, ammonia, and various petrochemicals, polymers, paints, and plastics. It is estimated that well over 50% of all the chemical products produced today are made with the use of catalysts.

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CHAPTER 19 THE PACKED BED CATALYTIC REACTOR

Figure 19.1 Various types of catalytic reactors.

© Page 47

Reactant gas can be made to contact solid catalyst in many ways, and each has its specific advantages and disadvantages. Figure 19.1 illustrates a number of these contacting patterns. These may be divided into two broad types, the fixed bed reactors of Fig. 19.1 a, b and c and the fluidized-bed reactors of Figs. 19.1 d, e, and f. The moving-bed reactor of Fig. 19.1g is an intermediate case which embodies some of the advantages and some of the disadvantages of fixed-bed and fluidized-bed reactors

The main difficulties of design of catalytic reactors reduce to the following two questions:

- (1) How do we account for the nonisothermal behavior of packed beds? and
- (2) How do we account for the nonideal flow of gas in fluidized beds.

Figure 19.1 (Continued.)

© Page 48

CHAPTER 22 G/L REACTIONS ON SOLID CATALYST: TRICKLE BEDS, SLURRY REACTORS, THREE PHASE FLUIDIZED BEDS

These multiphase reactions are of the type

and they can be run in a number of ways, as shown in Fig. 22.1. The packed bed contactors use large solid particles, the slurry reactors very fine suspended solids, while the fluidized bed can use either depending on the flow rates.

Flow pattern: Overall, all other things equal (which they are not) countercurrent plug flow is the most desirable flow pattern while the worst of all in terms of driving forces is mixed/mixed flow.

Figure 22.1 Various ways of running G/L reactions catalyzed by solids.

© Page 49

NON-CATALYTIC SYSTEMS (LEVENSPIEL RXN TEXTBOOK)

Chapter 23 Fluid-Fluid Reactions: Kinetics

Chapter 24 Fluid-Fluid Reactions: Design

Chapter 25 Fluid-Particle Reactions: Kinetics

Chapter 26 Fluid-Particle Reactions: Design

CHAPTER 23 FLUID-FLUID REACTIONS: KINETICS

The following factors will determine how we approach this process.

The Overall Rate Expression. Since materials in the two separate phases must contact each other before reaction can occur, both the mass transfer and the chemical rates will enter the overall rate expression.

Equilibrium Solubility. The solubility of the reacting components will limit their movement from phase to phase. This factor will certainly influence the form of the rate equation since it will determine whether the reaction takes place in one or both phases.

The Contacting Scheme. In gas-liquid systems semibatch and countercurrent contacting schemes predominate. In liquid-liquid systems mixed flow (mixer-settlers) and batch contacting are used in addition to counter and concurrent contacting.

THE RATE EQUATION

For convenience in notation let us talk of G/L reactions, even though what we say holds equally for L/L reactions. Further, let us assume that gaseous A is soluble in the liquid but that B does not enter the gas. Thus A must enter and move into the liquid phase before it can react, and reaction occurs in this phase alone.

Now the overall rate expression for the reaction will have to account for the mass transfer resistance (to bring reactants together) and the resistance of the chemical reactions step. Hence the rate equation for mass transfer and reaction needs to be considered.

© Page 50

CHAPTER 24 FLUID-FLUID REACTIONS: DESIGN

We must first choose the right kind of contactor, then find the size needed. There are two kinds of contactor-towers and tanks, and Fig. 24.1 shows some examples. As may be expected, these contactors have widely different G/L volume ratios, interfacial areas, k , and concentration driving forces. The particular properties of the system you are dealing with, the solubility of gaseous reactant, the concentration of reactants, etc. – in effect the location of the main resistance in the rate equation – will suggest that you use one class of contactor and not the other.

Figure 24.1 Tower and tank contactors for G/L reactions.

Contacting pattern: We idealize these as shown in Fig. 24.2.

- Towers approximate plug G/plug L.
- Bubble tanks approximate plug G/mixed L.
- Agitated tanks approximate mixed G/mixed L.

© Page 51

Figure 24.2 Contacting patterns for G/L contactors.

CHAPTER 25 FLUID-PARTICLE REACTIONS: KINETICS

This chapter treats the class of heterogeneous reactions in which a gas or liquid contacts a solid, reacts with it, and transforms it into product. Such reactions may be represented by

In this chapter we develop the rate expressions for fluid-solid reactions. The next chapter will then use this information in design.

SELECTION OF A MODEL

We should clearly understand that every conceptual picture or model for the progress of reaction comes with its mathematical representation, its rate equation. Consequently, if we choose a model we must accept its rate equation, and vice versa. If a model corresponds closely to what really takes place, then its rate expression will closely predict and describe the actual kinetics; if a model differs widely from reality, then its kinetic expressions will be useless. The requirement for a good engineering model is that it be the closest representation of reality which can be treated without too many mathematical complexities.

© Page 52

For the noncatalytic reaction of particles with surrounding fluid, we consider two simple idealized models, the progressive-conversion model and the shrinking unreacted-core model.

Progressive-Conversion Model (PCM). Here we visualize that reactant gas enters and reacts throughout the particle at all times, most likely at different rates at different locations within the particle. Thus, solid reactant is converted continuously and progressively throughout the particle as shown in Fig. 25.1.

Figure 25.1 According to the progressive-conversion model, reaction proceeds continuously throughout the solid particle.

Figure 25.2 According to the shrinking-core model, reaction proceeds at a narrow front which moves into the solid particle. Reactant is completely converted as the front passes by.

© Page 53

Shrinking-Core Model (SCM). Here we visualize that reaction occurs first at the outer skin of the particle. The zone of reaction then moves into the solid, leaving behind completely converted material and inert solid. We refer to these as "ash." Thus, at any time there exists an unreacted core of material which shrinks in size during reaction, as shown in Fig. 25.2.

Comparison of Models with Real Situations. In slicing and examining the cross section of partly reacted solid particles, we usually find unreacted solid material surrounded by a layer of ash. The boundary of this unreacted core may not always be as sharply defined as the model pictures it; nevertheless, evidence from a wide variety of situations indicates that in most cases the shrinking-core model (SCM) approximates real particles more closely than does the progressive conversion model (PCM). Observations with burning coal, wood, briquettes, and tightly wrapped newspapers also favor the shrinking-core model.

Difference between progressive conversion and the shrinking core model for fluid-solid non-catalytic reactions

Progressive Conversion Model Shrinking Core Model

In PCM, the reactant gas enters throughout In SCM, the reaction occurs first at the outer the particle at all times most likely at skin of the particle. The zone of reaction different location within the particle then moves into the solid, leaving behind completely converted material and inert solid

The solid reactant is converted continuously At any time, there exist an unreacted core of and progressively throughout the particle material which shrink in size during reaction

CHAPTER 26 FLUID-PARTICLE REACTIONS: DESIGN

Three factors control the design of a fluid-solid reactor;

- the reaction kinetics for single particles,
- the size distribution of solids being treated, and
- the flow patterns of solids and fluid in the reactor.

© Page 54

MULTIPLE REACTIONS

4.1 Single and Multiple (Complex) Reactions

When a single stoichiometric equation and single rate equation are chosen to represent the progress of the reaction, we have a single reaction. When more than one stoichiometric equation is chosen to represent the observed changes, then more than one kinetic expression is needed to follow the changing composition of all the reaction components, and we have multiple reactions.

A reaction that proceeds by a mechanism involving more than a single reaction path or step is termed a complex reaction. Unlike elementary reactions, the mechanisms of complex reactions differ considerably from their stoichiometric equations. Most industrially important reactions are complex reactions, the mechanisms of which can often be determined by assuming that the overall reaction consists of several elementary reaction steps. The resulting overall rate expression is then compared with the experimental data, and the procedure is repeated until a desired degree of agreement is obtained. Each of the elementary reaction steps may proceed reversibly, concurrently, or consecutively.

a) A reversible reaction is one in which conversion of reactants to products is incomplete at equilibrium because of an increasing influence of the reverse reaction as the forward reaction approaches equilibrium. For a reversible reaction of the type the net forward rate of reaction in terms of disappearance of A is

which assumes both forward and reverse reactions to be elementary

b) A simple example of consecutive or series reaction is illustrated by

$\rightarrow \rightarrow$

Again, assuming an elementary reaction for each reaction step, the following rate

equations result:

c) Parallel or simultaneous reactions are those involving one or more reactants undergoing reactions of more than one scheme, as in

© Page 55

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The rate equations may assume the following forms:

Under a constant-volume condition, $r_B = dC_B/dt$ and $r_D = dC_D/dt$. Therefore, the relative rate of formation of B and D is derived from Eqs. (4.2) and (4.3) as

The ratio dC_B/dC_D is termed the point selectivity, which is the ratio of the rate of formation of product B to the rate for product D. The overall (or integrated) selectivity is obtained by integration of this ratio, and it represents the ratio of the overall amount of product B to that of product D. Equation (4.4) implies that the relative rate of formation of B is proportional to C_A when $a > b$, whereas it is inversely proportional to C_A when $a < b$.

In a chemical process involving complex reactions that consist of several reaction steps, one or more steps may represent major factors in governing the overall rate of reaction. Such reaction steps are termed the rate-controlling steps. The rate-controlling reaction steps are observed in homogeneous complex reactions and heterogeneous reactions.

© Page 56

SOLUTION (Note CSTR 40% conversion & not PFR 70%)

© Page 57

© Page 58

© Page 59